Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/02/2016
PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	HZE Radiation Effects on Malignant Progression in Human Epithelial Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mary.barcellos-hoff@ucsf.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	415-476-8091
Organization Name:	University of California, San Francisco		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	2340 Sutter Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94115	**Congressional District:**	12
Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/23/2013	**End Date:**	01/22/2016
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AF06G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Little is known about the mechanisms that underlie the greater biological effectiveness of high linear energy transfer (LET) irradiation to promote solid cancer in experimental models. Our experimental data and that of others suggest that radiation carcinogenesis is a two-compartment problem: ionizing radiation can alter genomic sequence as a result of damage due to the interaction of energy and DNA, while altered phenotypes and multicellular interactions that may contribute to cancer remain poorly defined. The hallmarks of cancer occur within very specific microenvironments that promote malignancy (Hanahan and Weinberg, 2011). We hypothesize that both genetic and phenotypic changes contribute to radiation carcinogenesis. Radiation effects on critical phenotypes, if well understood, provide attractive targets for biological countermeasures because they represent processes that develop after radiation exposure and, unlike mutations, are amenable to intervention. We propose to study		

	three aspects of neoplasia: tissue stem cells, phenotype, and genomic instability as a function of HZE (high energy) particle irradiation of non-malignant epithelial cells co-cultured with irradiated human fibroblasts and/or macrophages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is focused on developing a new experimental model of early stages of breast carcinogenesis representing critical cell interactions during radiation carcinogenesis. We contend that a more detailed mechanistic understanding of cell interactions that are affected by radiation will help to reduce the uncertainties in the age and gender dependence of cancer risks from the space radiation environment. We use a systems biology approaches to integrate molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the relevance of the cancer model to human data. The premise for these studies, that a pre-cancer niche formed by stromal cells promotes survival and malignant progression of initiated epithelial cells, is a new paradigm in carcinogenesis conceptualized by the Principal Investigator. The concept was published in 2013 with colleagues in Nature Cancer Reviews. We propose that analysis of radiation effects on niche formation as a precursor to tumor microenvironment in this project will provide new data and a new perspective on preventing cancer. We hope to identify specific mechanisms that will serve as biomarkers that will allow comparisons among human tissues and will provide targets for countermeasures during space travel. Achieving these goals will also likely be useful to understanding and preventing risk from terrestrial radiation exposures.
	Background Cancer is an important long-term risk for astronauts exposed to protons and high energy charged particles during travel and residence on other planets and the moon. The mechanisms underlying the greater biological effect of carcinogenesis after HZE particle irradiation remain largely unknown. Our hypothesis is that increased cancer risk following radiation exposure results from the interaction of radiation targeted effects (TE), such as mutation, augmented by non-targeted effects (NTE) that affect cell phenotype and signaling (1,2). We model the tissue/organ/organism as a system rather than a collection of non-interacting or minimally interacting cells that can reveal emergent phenomenon due to system perturbation (2-5). Thus, cancer emerges in the context of host genetics and physiology as a result of a complex, but potentially predictable, interplay between radiation TE and NTE (2). To dissect the relative contributions of TE vs NTE in HZE carcinogenesis, we compared mammary tumorigenesis in mice irradiated with low fluences of different HZE particles when both the target epithelium and the host were irradiated (TE+NTE) compared to tumors arising from non-irradiated epithelium transplanted to irradiated hosts (NTE only). Notably, we identified a strong radiation quality effect on tumor aggressiveness, i.e., tumor features and behaviors associated with poor prognosis in humans, which is primarily a function of NTE. This published research show that HZE particle-irradiation acts via the microenvironment to promote aggressive tumors. We then irradiated mice aged to 10 months (approximately equivalent to the age of active duty astronauts) to evaluate the contribution of NTE to cancer risk in an age relevant context. This experiment revealed a previously unsuspected strong radiation quality effect: tumors arising from the aged host irradiated with HZE are much more aggressive, as defined by shortened latency, increased growth rate and intrinsic profiles, than tumors appearing in sham- or gamma-irradiated mice (Ouyang and Barcellos-Hoff, in preparation). Notably tumors arising in aged HZE-irradiated mice show a specific signature implicating corruption of anti-tumor immunity. Indeed, we now have extended this perspective based on in-depth analysis of several hundred tissues and tumors and postulate that the major risks of HZE radiation exposure, carcinogenesis, CNS (central nervous system) and degenerative pathologies, are linked by a unifying dependence on NTE mainly mediated by the immune system. The radiation quality and age dependence of NTE is arguably the most NASA relevant biology to be reported in the last decade of research. These concepts and new data provide a strong foundation to inform and reduce uncertainties in NASA radiation risk assessment. The data obtained from our prior studies of HZE radiation biology across levels of organization has led to new understanding of how complex organisms respond to radiation (reviewed in (5)). We generated significant fundamental knowledge of cellular, tissue, and organ response to HZE and their contribution to malignant processes. Using these data to understand how biology is affected by exposure to the space radiation environment serves as the foundation to address NASA research goals to assess the consequences of a complex space radiation environment using ‘omics’ and systems biology by interrogating complex databases now available for experimental models and humans aim to inform the biological basis for risk prediction. The questions remain as to how and to what extent radiation quality affects such processes is still poorly understood. The current funding was used to test specific biological mechanisms that could drive tumor aggressiveness that we observed in particle-irradiated mice (6). We established a model system of non-malignant human epithelial cells in which to test whether HZE irradiation altered epithelial phenotype and if so, whether immune cells augmented or amplified such responses as a function of radiation quality. These studies examined the stability of epithelial cell phenotype, i.e., behaviors that are the culmination of how the genome is expressed in cells. All cells have the same genome so differential expression of the genome is the basis for phenotypic diversity (e.g., between blood and brain cells or skin and bone). Phenotype can be plastic, that is a given cell type can revert to an earlier stage, or even a different cell type, in response to specific stimuli. We studied a phenotype called epithelial to mesenchymal transition (EMT), which normally occurs in early development stages and is characterized by a loss of epithelial polarity and cell-to-cell contacts accompanied by the acquisition of a mesenchymal and more motile phenotype. These features are also characteristics of one of the earliest events in the carcinogenesis process, dysplasia, which is the local loss of tissue architecture and cellular polarity that is often a precursor to cancer (7-11). Epithelial cells must also escape from anti-proliferative signals during cancer development. Transforming growth factor beta (TGFß) is thought to act as a tumor suppressor in large part because it suppresses epithelial proliferation (12). But TGFß is also a potent inducer of EMT; thus cells undergoing EMT-mediated dysplasia are also less proliferative, and epithelial cells must escape to the TGFß anti-proliferative effect for malignancy to progress (13). Finally, these cellular events do not occur in a vacuum; they occur in a tissue that is composed of many cells derived locally and distantly. Cells from the immune system include macrophages and lymphocytes that are both necessary to defend the organism from infections and from rogue cancer cells. Summary of Research under NNX13AF06G

Task Progress:

Cultured human epithelial cells were irradiated at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory analyzed using various assays for stem cell and EMT. Notably, both sparsely and densely ionizing radiation elicit phenotypic changes that are evidenced by increased plasticity and decreased tissue-specific architecture (7). Although the literature suggests that these processes are coupled (14,15), our computational and cell biology analysis showed that they are parallel independent processes.

However we noted an intriguing caveat--establishment of cell junctions prevented irradiated cells from exhibiting these features, yet if the irradiated cell sheet was disrupted, the program was readily evident. This observation provides important evidence that normal tissue structure, in which an epithelium forms a sheet, is a critical barrier to malignant disruption. Yet cancer does occur more frequently in irradiated mice and people; we then asked what might disrupt the epithelium in an actual tissue. We found a macrophage gene signature in analysis of irradiated tissues gene expression data indicates that macrophage phenotypes and distributions are affected by radiation (6,7,16). Thus we tested the hypothesis that distinct subtypes of differentiated and active macrophages act locally to release epithelial cells to undergo EMT and resistance to anti-proliferative signals.

To do so, we isolated precursor cells from normal human blood and generated distinct subtypes of macrophages. Monocytes are recruited from the blood to tissues where they become macrophages that conduct surveillance of tissues for infection, as well as clearing up dead cells, both of which require that they move about a tissue. To do so, they produce certain proteins that permit access by disruption of cell junctions. We found that differentiated subtypes of macrophages, but not un-differentiated macrophage precursors, cooperate with radiation-induced TGFß to induce EMT in normal mammary epithelial cells. Interestingly, although sparsely ionizing H+ or gamma-irradiated epithelial cells co-cultured with differentiated macrophages show some morphological changes and loss of the epithelial marker, complete EMT was only observed when epithelial cells were irradiated using Fe56 or Si28 particles. Moreover, growth factor resistant cells increased only in the in HZE irradiated monolayers cultured with macrophages.

Summary

The data from our experiments underscore the importance of the radiation quality and reveals that particle irradiation can exhibit unique effects compared to sparsely ionizing radiation.. Based on the cancer biology literature, such effects are plausible mechanisms that may contribute to the significant increase in aggressive cancer observed in our studies of epithelial carcinogenesis. Identifying different biological consequences of high versus low LET radiation may improve prediction risk models, as well as provide strategies for rational design of countermeasures.

References:

1. Barcellos-Hoff MH. Integrative radiation carcinogenesis: interactions between cell and tissue responses to DNA damage. Semin Cancer Biol 2005;15(2):138-48.

2. Barcellos-Hoff MH. Cancer as an Emergent Phenomenon in Systems Radiation Biology. Radiat Env Biophys 2007;47(1):33-8.

3. Barcellos-Hoff MH, Costes SV. A systems biology approach to multicellular and multi-generational radiation responses. Mutation Res 2006;597(1-2):32-38.

4. Barcellos-Hoff MH. What is the use of systems biology approaches in radiation biology? Health Phys 2011;100(3):272-3.

5. Barcellos-Hoff MH, Adams C, Balmain A, Costes SV, Demaria S, Illa-Bochaca I, et al. Systems biology perspectives on the carcinogenic potential of radiation. Journal of Radiation Research 2014;55(suppl 1):i145-i54.

6. Illa-Bochaca I, Ouyang H, Tang J, Sebastiano C, Mao J-H, Costes SV, et al. Densely Ionizing Radiation Acts via the Microenvironment to Promote Aggressive Trp53 Null Mammary Carcinomas. Cancer research 2014;74(23):7137-48.

7. Tang J, Fernandez-Garcia I, Vijayakumar S, Martinez-Ruiz H, Illa-Bochaca I, Nguyen DH, et al. Irradiation of juvenile, but not adult, mammary gland increases stem cell self-renewal and estrogen receptor negative tumors. Stem Cells 2013;32(3):649-61.

8. Andarawewa KL, Costes SV, Fernandez-Garcia I, Chou WS, Ravani SA, Park H, et al. Radiation Dose and Quality Dependence of Epithelial to Mesenchymal Transition (EMT) Mediated by Transforming Growth Factor ß. Int J Rad Onc Biol Phys 2011;79(5):1523-31.

9. Maxwell CA, Fleisch MC, Costes SV, Erickson AC, Boissiere A, Gupta R, et al. Targeted and nontargeted effects of ionizing radiation that impact genomic instability. Cancer research 2008;68(20):8304-11.

10. Andarawewa KL, Erickson AC, Chou WS, Costes SV, Gascard P, Mott JD, et al. Ionizing radiation predisposes nonmalignant human mammary epithelial cells to undergo transforming growth factor b induced epithelial to mesenchymal transition. Cancer research 2007;67:8662-70.

11. Park CC, Henshall-Powell RL, Erickson AC, Talhouk R, Parvin B, Bissell MJ, et al. Ionizing radiation induces heritable disruption of epithelial cell interactions. Proceedings of the National Academy of Sciences of the United States of America 2003;100(19):10728-33.

12. Silberstein GB, Daniel CW. Reversible inhibition of mammary gland growth by transforming growth factor-ß. Science (New York, NY) 1987;237:291-93.

13. Derynck R, Akhurst RJ, Balmain A. TGF-b signaling in tumor suppression and cancer progression. Nature Genet 2001;29:117-29.

14. Mani SA, Guo W, Liao M-J, Eaton EN, Ayyanan A, Zhou AY, et al. The Epithelial-Mesenchymal Transition Generates Cells with Properties of Stem Cells. Cell 2008;133(4):704-15.

15. Mani SA, Guo W, Liao MJ, Eaton EN, Ayyanan A, Zhou AY, et al. The epithelial-mesenchymal transition generates cells with properties of stem cells. Cell 2008;133(4):704-15.

16. Nguyen DH, Oketch-Rabah HA, Illa-Bochaca I, Geyer FC, Reis-Filho JS, Mao JH, et al. Radiation Acts on the Microenvironment to Affect Breast Carcinogenesis by Distinct Mechanisms that Decrease Cancer Latency and Affect

	Tumor Type. Cancer Cell 2011;19(5):640-51.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Gonzalez-Junca A, Lin M, Barcellos-Hoff MH. "HZE radiation promotes malignant progression of human epithelial cells by triggering an EMT program and selecting for TGFbeta-resistant cells." 2016 Human Research Program Investigators' Workshop, Galveston, TX, Feb 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Barcellos-Hoff MH. "Cancer Biology Working Group in Space Radiation." 2016 Human Research Program Investigators' Workshop, Galveston, TX, Feb 8-11, 2016. 2016 Human Research Program Investigators' Workshop, Galveston, TX, Feb 8-11, 2016. , Feb-2016
Articles in Peer-reviewed Journals	Ma L, Gonzalez-Junca A, Zheng Y, Ouyang H, Illa-Bochaca I, Horst KC, Krings G, Wang Y, Fernandez-Garcia I, Chou W, Barcellos-Hoff MH. "Inflammation mediates the development of aggressive breast cancer following radiotherapy." Clin Cancer Res 2021 Mar;27(6):1778–91. https://doi.org/10.1158/1078-0432.CCR-20-3215 ; PMID: 33402361 , Mar-2021
Articles in Peer-reviewed Journals	Decarlo L, Mestel C, Barcellos-Hoff MH, Schneider RJ. "Eukaryotic translation initiation factor 4E is a feed-forward translational coactivator of TGFß early pro-transforming events in breast epithelial cells." Mol Cell Biol. 2015 Aug;35(15):2597-609. Epub 2015 May 18. http://dx.doi.org/10.1128/MCB.00324-15 ; PubMed PMID: 25986608; PubMed Central PMCID: PMC4524117 , Aug-2015
Articles in Peer-reviewed Journals	Barcellos-Hoff MH, Mao JH. "HZE radiation non-targeted effects on the microenvironment that mediate mammary carcinogenesis." Front Oncol. 2016 Mar 11;6:57. eCollection 2016. http://dx.doi.org/10.3389/fonc.2016.00057 ; PubMed PMID: 27014632; PubMed Central PMCID: PMC4786544 , Mar-2016
Articles in Peer-reviewed Journals	Hendry JH, Niwa O, Barcellos-Hoff MH, Globus RK, Harrison JD, Martin MT, Seed TM, Shay JW, Story MD, Suzuki K, Yamashita S. "ICRP Publication 131: Stem cell biology with respect to carcinogenesis aspects of radiological protection." Ann ICRP. 2016 Jun;45(1 Suppl):239-52. Epub 2016 Mar 8. http://dx.doi.org/10.1177/0146645315621849 ; PubMed PMID: 26956677 , Jun-2016
Articles in Peer-reviewed Journals	Barcellos-Hoff MH, Blakely EA, Burma S, Fornace AJ Jr, Gerson S, Hlatky L, Kirsch DG, Luderer U, Shay J, Wang Y, Weil MM. "Concepts and challenges in cancer risk prediction for the space radiation environment." Life Sci Space Res (Amst). 2015 Jul;6:92-103. http://dx.doi.org/10.1016/j.lssr.2015.07.006 ; PMID: 26256633 , Jul-2015